

$$\int \frac{2x-1}{x(x+1)^2} dx = \int \frac{A}{x} dx + \int \frac{B}{x+1} dx + \int \frac{C}{(x+1)^2} dx$$

1. zabalzailea 3. mailako polinomio denez adierazten ariazonde 3 zatikitan deskonposatu daiteke.

Kasu honetan bi erro daude $x=0$ eta $x=-1$. Azken hau erro bikotza denez bi zatikira ere erabiltzeko dugu.

$$\frac{2x-1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

mk+(x.(x+1)²), beraz izen dabalda konur

$$\frac{(2x-1)}{x(x+1)^2} = \frac{A(x+1)^2 + Bx(x+1) + Cx}{x(x+1)^2}$$

zati bakoitza falka dene.

$$2x-1 = A(x+1)^2 + Bx(x+1) + Cx$$

$$x=0 \quad -1 = A \cdot 1 + B \cdot 0 + C \cdot 0 \quad A = -1$$

$$x=-1 \quad -3 = A \cdot 0 + B \cdot 0 - C \quad C = 3$$

$$x=1 \quad 1 = A \cdot 2^2 + B \cdot 2 + C$$

$$1 = 4A + 2B + C$$

$$1 = -4 + 2B + 3 \quad B = 1$$

$$\int \frac{2x-1}{x(x+1)^2} dx = \int \frac{-1}{x} dx + \int \frac{1}{x+1} dx + \int \frac{3}{(x+1)^2} dx =$$

$$= -\ln|x| + \ln|x+1| - \frac{3}{x+1} + K$$

$$a) \int \frac{5}{x^2 - 3x + 2} dx = \int \frac{5}{(x-2)(x-1)} dx = \int \frac{A}{x-2} dx + \int \frac{B}{x-1} dx$$

$$x^2 - 3x + 2 = 0 \text{ Faktorisasi}$$

$$x = \frac{3 \pm \sqrt{9-8}}{2} = \frac{3 \pm 1}{2} \quad \begin{matrix} x_1 = 2 \\ x_2 = 1 \end{matrix}$$

$$\frac{5}{(x-2)(x-1)} = \frac{A}{x-2} + \frac{B}{x-1}$$

zatk. bot bi zatukiren batuketa berala
apini

$$\text{mkt } (x-2)(x-1)$$

$$\frac{5}{(x-2)(x-1)} = \frac{A(x-1)}{(x-2)(x-1)} + \frac{B(x-2)}{(x-2)(x-1)}$$

$$5 = A(x-1) + B(x-2)$$

$$x=1 \quad 5 = A \cdot 0 - 1B \quad B = -5$$

$$x=2 \quad 5 = A + 0 \quad A = 5$$

$$\int \frac{5}{x^2 - 3x + 2} dx = \int \frac{5}{x-2} dx + \int \frac{-5}{x-1} dx = 5 \ln|x-2| - 5 \ln|x-1| + K$$

$$b) \int (2x+1)^4 dx \quad \textcircled{1} \text{ Ordekapen metodoa edo berehalakor } \textcircled{2}$$

$$\textcircled{1} \int (2x+1)^4 dx = \left\| \begin{matrix} t = 2x+1 \\ dt = 2dx \quad dx = \frac{dt}{2} \end{matrix} \right\| = \int t^4 \cdot \frac{dt}{2} = \frac{1}{2} \int t^4 dt =$$

$$= \frac{1}{2} \cdot \frac{t^5}{5} + K = \frac{1}{10} (2x+1)^5 + K$$

$$\textcircled{2} [f(x)]^4 \text{ daukagunet badekigu bere jatorrakor } f(x)^5 \text{ izan behar dela.}$$

$$[(2x+1)^5]' = 5(2x+1)^4 \cdot (2x+1)' = 10(2x+1)^4 \text{ izango lirateke, beraz.}$$

$$\int (2x+1)^4 dx = \frac{1}{10} (2x+1)^5 + K \quad \frac{(2x+1)^5}{10} = (2x+1)^4$$

$$\int_1^e x^2 \cdot \ln(x) dx =$$

ZATIKAKO INTEGRASIA erabiliz ebatzeko da.

Zatikako integrazioa biderketa baten deribatutik abiatuz garaturiko integrazioa da.

Biderketa baten deribatua $(u(x) \cdot v(x))' = u'(x) \cdot v(x) + u(x) \cdot v'(x)$ dela jakinik, honela adieraz daiteke: $d(u(x) \cdot v(x)) = du(x) \cdot v(x) + u(x) dv(x)$, ordenatuz.

$$u(x) \cdot dv(x) = d(u(x) \cdot v(x)) - du(x) \cdot v(x)$$

B. aldeetan integratuz:

$$\int u(x) \cdot dv(x) = \int d(u(x) \cdot v(x)) - \int v(x) \cdot du(x)$$

$$\int u(x) \cdot dv(x) = u(x)v(x) - \int v(x) du(x)$$

x-ekiko integralen deribatuak dugunez berberak

$$\int u dv = u \cdot v - \int v du$$

$\int x^2 \cdot \ln(x) dx$ integralerakoan, biderkagai bat u eta bestea dv

moduan hartu behar da, kasu honetan $u(x) = \ln(x)$ eta $dv = x^2 dx$

Beraz deribatuz

Integral defnitu a denez, badakigu Barrow-en erregela

aplikatu

$$\int_1^e f(x) dx = F(e) - F(1) \text{ izango da, beraz}$$

lehenengo jatorriak kalkulatu da eta gero mugako balioak ordezkatu.

$$\int x^2 \ln(x) dx = \left\| \begin{array}{l} u = \ln(x) \quad du = \frac{1}{x} dx \\ dv = x^2 dx \quad v = \frac{x^3}{3} \end{array} \right\| = \frac{x^3}{3} \ln(x) - \int \frac{x^3}{3} \cdot \frac{1}{x} dx =$$

$$= \frac{x^3}{3} \ln(x) - \frac{1}{3} \int x^2 dx = \frac{x^3}{3} \ln(x) - \frac{1}{3} \cdot \frac{x^3}{3} + k = \frac{x^3}{3} \ln(x) - \frac{x^3}{9} + k$$

$$\int_1^e x^2 \ln(x) = F(e) - F(1) = \left[\frac{x^3}{3} \ln x - \frac{x^3}{9} + k \right]_1^e =$$

$$= \left(\frac{e^3}{3} \ln(e) - \frac{e^3}{9} + k \right) - \left(\frac{1}{3} \ln(1) - \frac{1}{9} + k \right) =$$

$$= \left(\frac{e^3}{3} - \frac{e^3}{9} \right) - \left(0 - \frac{1}{9} \right) = \frac{2e^3}{9} + \frac{1}{9} = \frac{1}{9} (2e^3 + 1)$$

$$f(x) = x^2 - 2x + 1 \quad \text{Domain} = \mathbb{R}$$

$$g(x) = -x^2 + 5 \quad \text{Domain} = \mathbb{R}$$

AZALEKA

Esperma fungsi biak bat dorezen baloeek mupetuko dute:

$$x^2 - 2x + 1 = -x^2 + 5$$

$$2x^2 - 2x - 4 = 0 \quad \text{balokides} \quad x^2 - x - 2 = 0$$

$$x = \frac{1 \pm \sqrt{1+8}}{2} = \frac{1 \pm 3}{2} \quad \begin{cases} x_1 = 2 \\ x_2 = -1 \end{cases}$$

Indikabeko

$$f(x) = x^2 - 2x + 1 \quad (\text{Parabola}) \quad (\text{Anomza})$$

$$\text{Epina} \quad x_e = \frac{-b}{2a} = \frac{2}{2} = 1 \quad E(1, 0)$$

$$y_e = 1^2 - 2 + 1 = 0$$

↳ Balis tauba
x=1 en
ingwan

x	y
1	0
0	1
2	1 → *
-1	4 → *
3	4

$$g(x) = -x^2 + 5 \quad (\text{Parabola / Ganbika})$$

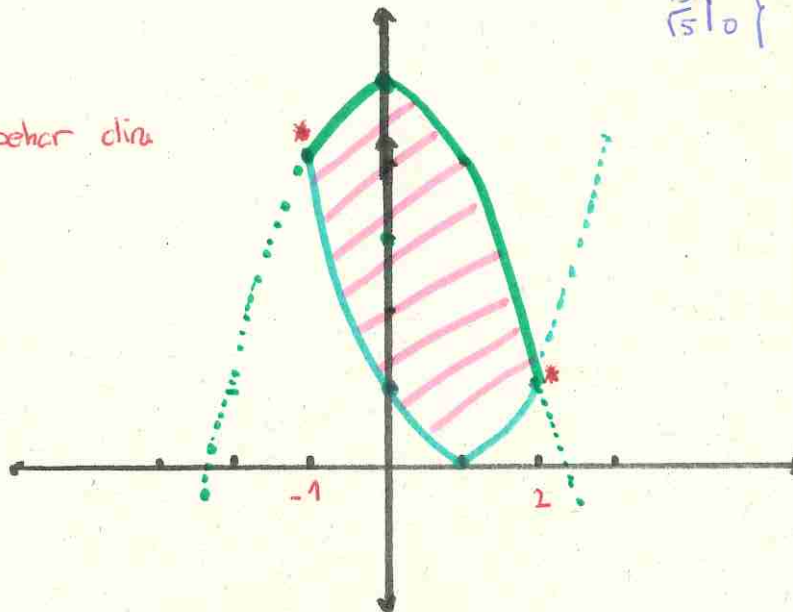
$$x_e = \frac{0}{2} = 0$$

$$g(0) = 5 \quad E(0, 5)$$

x	y
0	5
1	4
-1	4 → *
2	1 → *
-2	1
$\sqrt{5}$	0
$-\sqrt{5}$	0

x ordabaten ebeki puntuek
 $-x^2 + 5 = 0 \quad x = \pm\sqrt{5}$

* bat etorbekor dira



Azaler kalkulazeko kontuan eduki behar da zera funtzio
 mugitu duen eremua gortik eta zinez behetik. Kasu
 honetan $f(x)$ beheko mutura $g(x)$ gortiko mutura, bestela
 azaler negatiboa lortuko genuke

$$A = \int_{-1}^2 [g(x) - f(x)] dx = \int_{-1}^2 (-x^2 + 5) - (x^2 - 2x + 1) dx =$$

$\uparrow$ gortiko funtzio $\uparrow$ beheko funtzio $\uparrow$ Kontuz

$$= \int_{-1}^2 (-x^2 + 5 - x^2 + 2x - 1) dx = \int_{-1}^2 (-2x^2 + 2x + 4) dx =$$

$$= \left[-\frac{2x^3}{3} + x^2 + 4x \right]_{-1}^2 = \left(-\frac{2 \cdot 8}{3} + 4 + 8 \right) - \left(\frac{2(-1)}{3} + 1 - 4 \right) = \left(-\frac{16}{3} + 12 \right) - \left(-\frac{2}{3} - 3 \right)$$

$$= -\frac{18}{3} + 15 = -6 + 15 = 9 \text{ u}^2$$

Beronen ematen $A = \int_a^b f(x) dx = F(b) - F(a)$